

1(a)(i)	7 <u>nm</u> ; A a suitable range A within the range 5 to 10 <u>nm</u> <i>unit must be shown</i>	1																				
1(a)(ii)	one mark per row – name and letter must agree in row 3 – glycolipid and glycoprotein If no rows are correct or one row is correct, mark by column to give max 2 if one or two columns are correct <table><tr><th>component</th><th>function</th><th>letter on Fig.1.1</th><th></th></tr><tr><td>channel protein</td><td>facilitated diffusion or transport of, water / ions / water soluble substances / polar molecules / charged substances / hydrophilic substances I named examples of each category</td><td>A</td><td>;</td></tr><tr><td>phospholipid</td><td>forms a bilayer or a role within bilayer e.g. barrier to, water soluble / polar substances / ions allows diffusion of, fat soluble / non-polar, substances allows fluidity / provides stability / forms a hydrophobic core / AW</td><td>E</td><td>;</td></tr><tr><td>glycoprotein A glycolipid</td><td>receptor for cell signalling</td><td>B D</td><td>;</td></tr><tr><td>cholesterol</td><td>gives (mechanical) stability / maintains fluidity / regulates fluidity / barrier to water soluble substances or at low temperatures, maintains or increases fluidity / prevents close packing A prevents hydrophobic ‘tails’ interacting at low temperatures or at high temperatures, stabilises the membrane / decreases fluidity</td><td>F</td><td>;</td></tr></table>	component	function	letter on Fig.1.1		channel protein	facilitated diffusion or transport of, water / ions / water soluble substances / polar molecules / charged substances / hydrophilic substances I named examples of each category	A	;	phospholipid	forms a bilayer or a role within bilayer e.g. barrier to, water soluble / polar substances / ions allows diffusion of, fat soluble / non-polar, substances allows fluidity / provides stability / forms a hydrophobic core / AW	E	;	glycoprotein A glycolipid	receptor for cell signalling	B D	;	cholesterol	gives (mechanical) stability / maintains fluidity / regulates fluidity / barrier to water soluble substances or at low temperatures, maintains or increases fluidity / prevents close packing A prevents hydrophobic ‘tails’ interacting at low temperatures or at high temperatures, stabilises the membrane / decreases fluidity	F	;	4
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Question	Answer	Marks
1(b)(i)	<i>each organelle must be drawn in the cytoplasm and labelled for a mark</i> <i>max 2 if organelles are drawn correctly but not labelled or are not labelled correctly</i> <i>labels must clearly identify the organelles</i> R any 3D drawings mitochondrion with two membranes and at least one crista ; rough endoplasmic reticulum with at least one cisterna (two lines close together) and ribosomes attached on the outside ; R if any ribosomes completely inside the organelle smooth endoplasmic reticulum with at least one tube with no ribosomes ; A RER and SER as labels A if RER or SER attached to nuclear envelope	3
1(b)(ii)	Golgi body / dictyosome ; A Golgi, apparatus / complex <i>plus any one from:</i> modifies / processes, protein(s) / polypeptide(s) / lipid(s) ; A any suitable type of modification or a description e.g. glycosylation / addition of sugars forming tertiary or quaternary structures packaging of proteins into (Golgi) vesicles ; I transports AVP ; e.g. forming lysosomes / cell wall enzymes assembled	2